Sharvari Deshpande

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Education

Purdue University, John Martinsons Honors College West Lafayette, IN
Bachelor of Engineering (BE), Computer Engineering *Aug. 2023 – May 2027*

- 3.5 GPA, Dean's List, Semester Honors
- Minor in Math, Statistics, Certificate in Entrepreneurship and Innovation
- Undergraduate Teaching Assistant for ENGR 132: MATLAB Fundamentals (Spring 2025)
- Relevant Coursework: Data Structures and Algorithms, Python for Data Science, Artificial Intelligence, Linear Algebra, Introduction to Probability

Experience

Undergraduate Research Assistant Jan. 2026 - Present
Purdue Human Motor Behavior Group *West Lafayette, IN*

- Developed step detection algorithms in Python and MATLAB using Fourier analysis, signal filtering, and ML feature engineering via processing 1M+ sensor data points across experimental cycles.
- Reviewed and synthesized academic literature on computational gait analysis to brainstorm algorithm design.
- Presented weekly findings to research mentors and fellow peers.

Engineering Intern Jun. 2025 - Aug. 2025
Hawaiian Electric Company, Inc. *Hilo, HI*

- Applied ML methods such as Ward's linkage clustering and Dynamic Time Warping in Python, SQL, and Apache Spark (Databricks) to raw voltage data, uncovering overvoltage trends to enhance monitoring.
- Led RTU replacement efforts at Kawailani Electrical Substation by performing circuit analysis and editing CAD wiring diagrams.
- Collaborated with engineers and contractors to monitor progress and manage project timelines.

NASA S.U.I.T.S. Team Jarvis Member Aug. 2023 – Present
Space and Earth Analogs Research Chapter of Purdue (Purdue SEARCH) *West Lafayette, IN*

- Developed an interactive astronaut interface in C# on UNITY, specializing in simulating task management for NASA's Artemis missions.
- Co-authored a proposal for an autonomous rover and AR astronaut interface, contributing to pathfinding algorithms and LiDAR or camera-based localization methods.
- Team received the "Pay It Forward" Award for contributions and STEM outreach. Honored with a NASA Artemis I mission flag that orbited the Moon aboard the Artemis I rocket.

Undergraduate Data Science Researcher, Viasat Inc. Aug. 2024 – Jan. 2025
The Data Mine - Purdue University *West Lafayette, IN*

- Co-performed data ETL operations using Python and R to collect and clean satellite data from DiscosWeb API for advanced geospatial analysis.
- Co-presented critical satellite data findings to Viasat Inc. company representatives.

Projects

TruthDrift: AI Hallucination Detector Apr. 2026 – May 2026

- Developed an ML model using Python libraries utilizing features such as TF-IDF, Jaccard Similarity, Cosine Similarity, and GloVe embeddings to identify AI generated text hallucinations.
- Developed a Streamlit UI with per-sentence predictions and confidence scores for user inputted AI generated text.
- Improved model accuracy by around 10% via removing low-value GloVe embeddings during future iterations.

EWI: Electronic Wind Instrument Mar. 2026 – May 2026

- Captained a team to design & build a fully functional electronic wind instrument on the RP2350 microcontroller.
- Implemented tremolo in embedded C via I2C to an AD5693 DAC, with GPIO button control.
- Programmed & wired SPI-driven LCD display to render "EWI" in custom pixel art lettering.

Technical Skills

Languages: Python (Pandas, NumPy, Matplotlib, Flask, scikit-learn, Streamlit), JavaScript, SQL, R, C, C++, C#, HTML/CSS
Systems & Hardware Design: Verilog (RTL), embedded C, Assembly (RISC-V), Finite State Machines (FSMs), SPICE tools (LTspice, PSpice), FPGA development, KiCAD, Embedded Systems (RP2350 or Raspberry Pi Pico 2)
Developer Tools: Git, GitHub, Databricks, PlatformIO, Apache Spark (PySpark), UNITY, Visual Studio Code, Linux, Excel, MATLAB, Google Colab, Jira, LaTeX, Azure DevOps